Bioplastics interactions with microbial communities along the fresh and saltwater interface

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First PhD Planning Report

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1. **Table of content**

2. **Background**…….1

2.1. Introduction……1

2.2. Plastic Ecological Impact……..2

2.3. Bioplastics……..3

2.3.1. Definition, types, production, uses and trends……3

2.3.2. Environmental impact…….5

2.3.3. Biodegradation……...……6

2.4. Microbial communities and plastic…….7

2.4.1. In estuaries……..……7

2.4.2. Microbiological processes for transport and fate……8

2.4.3. Biodegradation and plastic degrading microorganisms ……9

2.4.4. Biogeochemical functions……11

2.4.5. A dispersal vector…….……12

3. **Aims and objectives of the project**……14

4. **Initial experimental/research design**……14

4.1. Research approach……14

4.2. Methods……17

4.3. Research techniques……..……...19

5. **Data analysis indications**……20

2. Background

2.1 Introduction

Plastics have been introduced a 100 years ago with the first real synthetic, mass-produced plastic, the Bakelite. Their extraordinary large-scale production and use however, started after World War II when it shifted from being mainly military to being used in almost every aspect of the modern life – food packaging, electronics, transports, building and construction, healthcare, agriculture, clothing to name a few (*PlasticsEurope*, 2019; Bucknall, 2020).

The term "plastic" is used to describe synthetic materials made from organic polymeric substances, commonly derived from petrochemicals, and of large molecular weight. In fact, approximatively 8% of the world's crude-oil is currently used to produce plastics. Although there are thousands of different plastic forms commercially available, only seven account for 70-90% of the total demand: low and high-density polyethylene (LDPE and HDPE), polypropylene, (PP), polyvinyl chloride (PVC), polystyrene (PS), polyethylene (PE) and polyethylene terephthalate (PET) (*PlasticsEurope*, 2019; Dilkes-Hoffman *et al*., 2019). They have many applications and positive characteristics such as being mouldable, flexible, lightweight, cheap and most of all durable (Dilkes-Hoffman *et al*., 2019). Packaging can increase food shelf-life and thus reduce food waste. Being light weight and cheap means that more can be transported and reduce the amount of greenhouse gases emitted by transportation of goods (Bucknall, 2020). In the past 70 years, their production increased exponentially and it is now estimated that over 9,342 billion metric tons of plastics have been produced as of 2019 (combining data from (*PlasticsEurope*, 2018; *PlasticsEurope*, 2019; Geyer, Jambeck and Law, 2017)).

However, due to improper disposal, a considerable amount of plastic waste is being produced. Geyer *et al* calculated that as of 2015, 76% of plastic produced generated plastic waste. 21% of which was recycled or incinerated and 79% accumulated in landfills or the natural

environment. This means that 60% of plastic being produced will eventually turn into environmental pollutant, with some inevitably entering aquatic environments. When projecting current trends, 12 billion tons of plastics waste would be discarded in landfills and polluting the natural environment. With 11 million metric tons of plastic flowing into the ocean every year and expected to reach 29 million tons in 2040 as studies predict (SYSTEMIQ - Breaking the Plastic Wave full report, 2020; Geyer, Jambeck and Law, 2017).

2.2. Plastic ecological impact

Herein, their biggest advantage, being durable, also becomes their biggest problem. Plastics build up in the environment as their durability outweighs their intended service life.

Plastic waste accumulating in the environment cannot naturally degrade but undergo physical, chemical and biological processes (i.e: UV exposure, mechanical abrasion) and decompose into smaller fragments forming micro (0.1 μm–5 mm) and even nano (<0.1 μm) particles (Alimba and Faggio, 2019; Shen, Song, *et al*., 2020). Such small sizes further their transportation potential meaning that plastic are increasingly being observable in all components of most aquatic ecosystems across the world, in terrestrial and marine environments, from the equator to the poles (*PlasticsEurope*-report, 2019; Barnes *et al*., 2009; Alimba and Faggio, 2019; Dilkes-Hoffman *et al*., 2019).

Numerous studies have demonstrated how plastic's ubiquity poses a number of physical, biological and toxicological threats to freshwater and marine biota such as entanglement, ingestion, the potential dispersal of invasive species and pathogens, additive and plasticizer toxicity, contamination through trophic levels (NOAA-MDP, reports 2019; Rochman *et al*., 2016; Galloway, Cole and Lewis, 2017; Law, 2017; Nagalakshmaiah *et al*., 2019; Ostle *et al*., 2019). Another major concern is the associated greenhouse gas footprint associated with plastic production (Dilkes-Hoffman *et al*., 2019) and the potential impact on Ocean Carbon

Sequestration has also been raised (Shen, Ye, *et al*., 2020). Plastic has now also become a societal and economical problem, with a resulting predicted excess cost of $13billion each year (Nizzetto *et al*., 2016; Agamuthu *et al*., 2019; Beaumont *et al*., 2019).

2.3. Bioplastics:

2.3.1. Definition, types, production, uses and trends

Emerged in the early 1980s, bioplastics are being developed among the many solutions proposed by a circular economy model to address the plastic problems caused by durable plastics being inappropriately disposed of (Bucknall, 2020; Jeremic *et al*., 2020).

There are some common confusion with the term "bioplastic" by the public but also in academic literature. The term bioplastic is often used for either bio-based plastics, biodegradable plastics or both. This common mistake can induce in error since bio-based plastics can be non-degradable and all biodegradable plastics are not necessarily bio-based. When bio-based answers to problematic feedstock, biodegradable supposedly addresses the problematic fate of mismanaged traditional plastic (TP) waste. Ultimately, there are 3 types of bioplastics (Table 1). Hereafter, the distinction will be made between biodegradable plastics and bio-based plastics. Biobased Biodegradable plastics (BBP) will refer to bioplastics that are biodegradable and bio-based conjointly and "bioplastics" will refer to all 3 types (Table 1).

Certain TP such as PET or PE are now producible from green, renewable feedstock. Such plastics conserve the same properties and molecular structure to petrochemical polymers and thus, are not biodegradable and need to be recycled (Lambert and Wagner, 2017).

Oil-based plastics described as biodegradable, are TPs such as HDPE, PCL or oxo-plastics, which include additives facilitating their breakdown and therefore, accelerate and often programme their fragmentation (i.e.: through oxidation, UV exposition). These plastics

however, aren't necessarily compostable and it is not known if they are fully biodegradable nor if they do so at a reasonable rate (Union, 2016; Lambert and Wagner, 2017).

Then there are the biodegradable bioplastics made from renewable resources (BBPs), which are expected to be the most promising alternative to TP as they appear to be of low-ecological impact and from sustainable, lower energy use, resources.

Table 1. Three types of bioplastics. Adapted from (Iwata, 2015; Kjeldsen et al., 2018; Jeremic et al., 2020)

	Bio-based	Oil-based
Biodegradable	Polylactic acid (PLA) Polyhydroxyalkanoate (PHA) Cellulose esters Bio-PBS Polysaccharide derivative (Starch blends) PBAT	Polylcaprolactone (PCL) Polybutyrate adipate terephthalate (PBAT) Polybutylene succinate (PBS)
Non-biodegradable	Bio-PE Bio-PET Bio-PP Polysaccharide derivative (starch blends)	Polyethylene (PE), Polypropylene (PP), Polystyrene (PS), Polyethylene terephthalate (PET), Polyvinyl chloride (PVC), Low and High density Polyethylene (LDPE, HDPE)

The most common BBPs, based on global production, are Starch blends, Polylactic acid (PLA), polybutylene adipate terephthalate (PBAT) and the polyhydroxyalkanoates (PHAs). With 2.11 million tons of bioplastics manufactured in 2019, their production respectively represents 21%, 14%, 13% and 1,2% of the lobal production of bioplastics. Although the range of BBPs applications is wide, going from packaging, food service, agriculture, electronics, automotive and consumer goods and household appliances, the demand for packaging remains the highest with 53% of the global production (1.14 million tons) (European Bioplastics - report, 2019; Jeremic *et al*., 2020)

Given that production has been increasing exponentially since their development representing 1% of the plastic being produced annually, it is expected that their input into aquatic environments intensifies too (European Bioplastics - report, 2019).

To sum up, BBP appear to be a promising part of the solution to the plastic waste problem. As it appears, they are sustainably derived and biodegradable, should not accumulate in the environment, they decouple plastics from the oil industry and limited petroleum resources and have a lower carbon footprint. However, in some cases, they will have inferior technical properties, be more expensive but most importantly, their fate in natural environment and ecological impact remain realistically largely unknown. In effect, claims that bioplastics have a lower environmental impact lacks of sufficient evidence (Walker and Rothman, 2020)..

2.3.2. Environmental impact

It is not known whether these new polymers degrade completely in nature, how they degrade and to what extent. Furthermore, there is growing concern that certain BBP may have similar impacts than TPs in the natural environment or have a worse energy use production (Bucknall, 2020). Effectively, in a 2020 review comparing the Life Cycle Assessment of 50 bio-based polymers with 39 TP studies, Walker and Rothman found that ecotoxicity and acidification showed far worse result for bio-based plastics (Bio-PET) than the worse TPs. PLA and other bioplastics showed worse eutrophication than fossil-fuel polymers. PLA display energy uses of 40-90MJ/kg of polymer as opposed to around 80MJ/kg TP polymer (Walker and Rothman, 2020). Another recent comparative study by Zimmermann et al (2020) on the toxicity and chemical composition of BBPs and TP actually found similar toxicity and amount of chemicals in BBPs and TPs (Zimmermann *et al*., 2020). Conversely, most studies on bioplastics are very recent and lack in standardisation for the methodology used to assess their ecological impact. Moreover, this great variation in methodology and the fact that studies on BBP are very recent compared to studies on TP, means that the variation between studies assessing the ecological impact of TP and BBPs doesn't easily allow for their comparison and is therefore impossible

to know if the variation between studies results indicate a genuine variation or whether the variation is cause by the variation in methods used (Walker and Rothman, 2020).

2.3.3. Biodegradation

There is also confusion in the general public around the distinction between what is degradable, biodegradable and compostable. Degradable generally refers to materials undergoing significant structural change in a definite environment whereas biodegradation refers to the breakdown of plastic by biological activity and undergo significant mass lowering. Generally, there are two steps to the biodegradation. The first one is the biophysical degradation, the fragmentation from high to low polymer by oxidation, photodegradation or hydrolysis and where the molecular structure remain unchanged. Then comes step two, where the fragmented polymer is biologically assimilated and mineralised by microorganisms and enzymes into water, biomass and carbon dioxide (under aerobic conditions) and methane (under anaerobic conditions) (Amaral-Zettler, Zettler and Mincer, 2020; Jeremic *et al*., 2020; Zhu and Wang, 2020). As for compostable, it refers to a material capable of biodegradation under specific compost conditions, at a rate consistent with other known compostable materials and without leaving visually distinguishable or toxic residues (Vert *et al*., 2012; Emadian, Onay and Demirel, 2017; Kjeldsen *et al*., 2018; Jeremic *et al*., 2020)

Claims about the biodegradability and compostability of some BBP are controversial. This is because the biodegradability of a material depends on many different environmental factors (biotic: microbial population richness, and abiotic: pH, sunlight, temperature, salinity, abrasion), polymer chemistry and the types and number of additives and plasticizers used (Zhu and Wang, 2020). To this point, most studies on bioplastic biodegradation have been conducted under very specific, controlled, laboratory environments, conditions and don't reflect real life conditions (Kjeldsen *et al*., 2018). Consequently, most bioplastics won't actually biodegrade

in all environments and under just any conditions. Furthermore, "The method of determining biodegradability is largely reliant of the development of gas from the material over time. This method may not take into account the environmental fate of any other components of the plastic, such as additives, or final polymeric components that cannot be further biodegraded." (Kjeldsen *et al*., 2018) . Even though the 138 bioplastics are either made from plants or have the ability to biodegrade under specific conditions, they are still plastics and are not necessarily "planet Safe" (Jeremic *et al*., 2020).

Herein, some BBPs such as PLA which is considered as biodegradable, hardly degrades in terrestrial and aquatic environments and compost under very specific, industrial, conditions (Lambert and Wagner, 2017; Jeremic *et al*., 2020). In a study, Vannini et al (2020) looked at PHBV and PHBV-OP in two different environments mesocosms (sand dunes and seabed sediments types) and found significantly different biodegradation rates between the two habitats. Besides, much higher degree of degradation were observed for the same bioplastics after being immersed in coastal tropical waters (Imam *et al*., 1999).

To understand BBPs ecological impacts, it is important to understand the physical processes affecting their fate into the environment, their trophic transfer and their chemical biodegradation but also what happens from a microbiological perspective.

2.4. Microbial communities and plastics

2.4.1 In estuaries

Estuarine are especially important when looking at plastic pollution impact. These environments are considered to be buffer zones between freshwater and saltwater and have unique ecological functions. They also represent areas of intense accumulation of plastic pollution. In effect, rivers constitute one of the main source for plastic discharge into the ocean with an estimated 1.15-2.41M tonnes each year (Lebreton *et al*., 2017; Alimba and Faggio,

2019). Additionally, being one of the most valuable ecosystems of the world for the variety of goods and services they provide to humans, urban and economic centres tend to thrive in these areas (Costanza *et al*., 1997; Browne, Galloway and Thompson, 2010). As a result, they are intensively impacted by human activities. Plastic inputs by the population is expected to be significant in these areas as it closely correlates to population size (Browne *et al*., 2011; Yonkos *et al*., 2014).

BBP and TP in aquatic environments are rapidly colonised by microorganisms and successively form biofilms (Dang and Charles R. Lovell, 2016; Abed *et al*., 2020). When this surface is in fact a plastic, the microbial community (MC) associated is called "plastisphere" (Amaral-Zettler, Zettler and Mincer, 2020). This biofouling is likely operating rapidly during the residence time in estuaries. Evidence of temporal shifts have previously been documented (Eich *et al*., 2015; Oberbeckmann, Osborn and Duhaime, 2016; Dussud *et al*., 2018a) and specific assemblages successions have been observed on plastics in coastal marine sediments and seawater over periods of 2 to 3 weeks only (Harrison *et al*., 2018). It was also observed that biofilm growth rate and its bacterial diversity were primarily affected by salinity and nutrient load (Li *et al*., 2019).

2.4.2. Microbiological processes for plastic transport and fate

It remains unclear whether these ecosystems act as a sink or source of plastic into the ocean, nor which location within estuaries are more prone for TP accumulation. This is because sorting and fate of TP depend on many physical, chemical and biological mechanisms such as, the type of estuary (salt wedge or well mixed), the effect of wind, salinity gradient, relative balance of fresh and saltwater flows, plastic type and temporal dynamic (seasonal changes, storms) to name a few. But most importantly, biodegradation and biofouling (Vermeiren, C. Munoz and Ikejima, 2016). Density plays a big role determining TP position in the water column and their

transport into the environment. Studies have shown how plastics biofouling and biodegradation may affect their densities and hydraulic properties and therefore, alter their transport and fate into the environment (Chen *et al*., 2019; Wu, Zhang, Li, *et al*., 2020). Herein, a plastic able to accumulate substantially higher amounts of microbial communities, with a large surface to volume ratio, will likely sink faster (Long *et al*., 2015; Ryan, 2015). Furthermore, plastic sinking rate might be further increased by degradation and biofilm formation facilitating the attachment of organic and inorganic materials (Long *et al*., 2015; Vermeiren, C. Munoz and Ikejima, 2016). Whereas biodegradation itself might also increase plastic buoyancy and cause the plastic to resurface. Therefore, density and diversity of the microbiota is an important factor for plastic buoyancy (Ryan, 2015). Subsequently, TP might undergo a series of sinking and resurfacing phases in the water column. This "yo-yo" effect has been described a number of times in marine settings but not detailed in estuarine conditions, especially for BBP.

For instance, it is not known how TP behave along the salinity gradient although a number of physical and chemical changes occur along this gradient (Telesh and Khlebovich, 2010). Furthermore, such processes do not act alone in determining plastic distribution at a whole estuary scale. An integrative understanding of the physical, chemical and biological mechanisms driving plastic sorting is therefore needed to understand the location and extent of source and sinks. The exact nature of connecting mechanisms need to be validated within estuarine contexts along the salinity gradient but also at different UV light exposition and oxygen levels. (Vermeiren, C. Munoz and Ikejima, 2016)

2.4.3. Biodegradation and plastic degrading microorganisms

The density and diversity of the biofilm is an important factor for buoyancy because it directly act on TP and BBP biodegradability. The biological processes to biodegrade plastics require specific bacteria. To date, over 90 types of microorganisms have been identified to degrade

bioplastics. Nonetheless, those are species commonly found in soil and compost materials and the biodiversity of BBP-degrading microorganism is not the same under different environments (Lee, Gimore and Huss, 2005; Kumaravel, Hema and Lakshmi, 2010; Accinelli *et al*., 2012; Morohoshi *et al*., 2018; Abed *et al*., 2020). For example, it was recurrently observe that soil and compost contained high numbers of bioplastic-degrading microorganisms because their microbial diversity is higher than that of seawater (Emadian, Onay and Demirel, 2017). Additionally, it was reported that soil microorganisms capable of degrading bioplastics were composed of bacteria and fungi species whereas bacteria species were the main microorganism found for aquatic environments (Emadian, Onay and Demirel, 2017).

Eich et al (2015) found that the microbial structure changes over time, with different marine habitats and differ for BBPs compared to TP. Moreover, Abel et al. (2020) found two bacteria genra which were specific to OXO-PE, a petrochemical biodegradable plastic, suggesting their possible involvement in its biodegradation. It is not known however if those bacteria will degrade other types of bioplastics.

Furthermore, it is important to consider the active genes of such bacteria since the solely presence of a plastic-degrading bacteria on a plastic might not mean it is actively degrading such plastic. In effect, intra- and extracellular enzymes such as depolymerases, lipases, estrases and serines have been identified as degrading specific types of bioplastics. Biodegradation of bioplastics happen by the action of enzymes of chemical destruction coming from bacteria, fungi or algae (Zhu and Wang, 2020). Yet again, these studies involve very few types of bioplastics and none have been conducted along the estuary gradients (Tokiwa and Calabia, 2004; Chua, Tseng and Yang, 2013; Trivedi *et al*., 2016; Morohoshi *et al*., 2018). Among the studies conducted to investigate the biodegradability of bioplastics in the environment, particular attention were given to terrestrial and marine environments but none on estuary environments (Emadian, Onay and Demirel, 2017).

2.4.4. Biogeochemical functions

In addition, this colonisation by microorganisms and subsequent biofilm formation on TP and BBP appear distinct compared to free-living communities (Oberbeckmann, Osborn and Duhaime, 2016). Surface MC are usually more diverse, dynamic (seasonal compositional variation), from larger morphologies, enriched and possess different functions than their free-living counterparts. (Dang and Charles R. Lovell, 2016). Consequently, surfaces are considered to be "hot spots" for microbial activities and support diverse and elevated biogeochemical activities. Surface associated marine microbiota participate in a plethora of Carbon, Nitrogen, Sulfur and iron cycling (Dang and Charles R. Lovell, 2016). Therefore, attached MC might differ in biogeochemical functions than surrounding or suspended microbes.

Microorganisms have the ability to bio-transform plastic debris into compounds that could pose a risk to human health and food security (McCormick, 2011). Facilitating the transport of MC with foreign biogeochemical functions into new ecosystems. "Surface associated microorganisms play important roles in numerous critical marine processes, including organic matter remineralisation and nutrient regeneration and element cycling" (Dang and Charles R Lovell, 2016).

For example, in 1994, Doering *et al*. described the potential stimulation of denitrification when estuarine sediments were treated with starch BBP and bacterial polyester (PHBV). This could be particularly detrimental in oligotrophic environments but could also help diminish nitrogen load in eutrophic estuaries. More recently, it was found that BBP might affect benthic biogeochemical activities through the stimulation of sulfate-reducers compared to TPs (Pinnell and Turner, 2019). Now, among the consequences, stimulating sulfate reducers is believed to play a major role in Accelerated Low Water Corrosion (Dang and Charles R Lovell, 2016). If BBPs are similarly discarded as TPs, the subsequent BBP waste entering aquatic system will

introduce a new surface for microbial colonisation. And it is not known yet which biogeochemical functions their MC could have, especially in an estuarine context.

2.4.5. A dispersal vectors

In their meta-analysis, Beaumont *et al*., (2019) highlighted that TP increased the range of habitats available for colonisation and enabled the spread of these species to new areas, thus increasing their range and abundance. Microplastics increase the ecological risk and probability of species invasion (Wu, Zhang, Zhao, *et al*., 2020). Consequently, microbial association to TP and BBP might exert the deleterious effect of persistence and transmission of harmful or pathogenic microorganisms and virulence determinants (Dang and Charles R. Lovell, 2016). Thanks to the durability of plastics in water, they provide a substrate that can be colonised by pathogens and harmful algal bloom species that can be transported long distances.

The predatory ciliate Ephelota, which can be an animal ectoparasite (Stankovic *et al*., 2002) or eat ciliates and other small eukaryotes, has commonly been found on TP samples, from marine or freshwater environment. In addition to invasive microorganisms, the colonisation by various potential pathogens have also been reported on bioplastics in aquatic systems around the world, including estuaries. Such as members of the Campylobacteraceae, *Aeromonas salmonicida* or *Arcobacter* spp., or members the genus *Vibro* which can cause diseases for wildlife and humans (Amaral-Zettler, Zettler and Mincer, 2020; Laverty *et al*., 2020). Additionally, the problematic propagation of pathogens is being coupled by the emergence and spread of antibiotic resistance genes (ARG), also considered to be an emerging environmental pollutant (Pruden *et al*., 2006). Not only such pollutant has been found in coastal water and sediments (Lu *et al*., 2015; Guo *et al*., 2018), they have also been found in higher abundances on TP biofilms compared to in water and sediments in estuary areas (Guo *et al*., 2020). Types of TP also appeared to be a factor influencing selectivity by antibiotic resistant bacteria and salinity to be an important

environmental regulator for the distribution of ARGs (Guo *et al*., 2018). It is therefore important to investigate the potential for BBP to carry and accumulate special bacteria in an estuary.

To recapitulate, there is growing evidence that BBP might impact the natural environment just as much as TP. From a microbiological perspective, BBP could alter ecosystems biogeochemical functions, transport predators, pathogens and other special microorganisms long distance. Furthermore, microbiological processes might affect their transport and fate into the environment, and it is not known whether they will actually biodegrade under any conditions. Whilst most studies on microbial communities have been conducted on TP, very few have been realised on BBP. Moreover, understanding their interaction in estuaries is close to inexistant. Filing those gaps in knowledge are of paramount urgency as TP have already began being replaced by BBP. This will help future management, production and creation of BBP.

3. Aims and Objectives of the project

Aim: Understand BBPs interaction with microbial communities along the fresh/saltwater gradient to explain their transport and fate and determine their associated biogeochemical functions and special bacteria colonisation, in comparison with TPs.

Objective 1 (O1): Increase a baseline of the microbial communities' diversity and distribution associated with TP considering environmental physicochemical factors within estuaries.

Objective 2 (O2): Determine microbial communities and BBP interaction considering bioplastic types and environmental physicochemical factors.

Objective 3 (O3): Understand how microbiological processes determine BBP and TP transport and fate.

Objective 4 (O4): Understand BBPs and TPs microbial communities' structure, dynamic and biogeochemical function.

Objective 5 (O5): Evaluate BBP biodegradation and determine diversity and distribution of bioplastic-degrading populations.

Objective 6 (O6): Characterise the special microbiota (predators, pathogens, ARGs) associated with BBP and their frequency and abundance.

4. Initial experimental/research design

4.1. Research approach

In situ

Most of the available studies assessing the interactions between MC and BBP, or even TPs, have been conducted in standardized laboratory conditions and over limited periods of time (Jacquin *et al*., 2019; Amaral-Zettler, Zettler and Mincer, 2020). Whilst this is good, there is a need for real life conditions studies and field testing in order to gain a clear picture of the processes really undergoing in nature.

Most importantly, the problem with micro or mesocosms experiments when studying bacterial assemblages is that they reflect colonisation that occurred only within those closed systems. It has been found in some studies that MC richness was lower on plastic than in surrounding areas, but most studies have been done only in laboratory settings or mesocosms. This can give inconclusive results with regards to plastics acting as dispersal vectors, or the potential of special bacteria accumulation. Especially because certain studies has revealed the potential for bioplastics to be colonized by bacteria during their transportations (Vannini *et al*., 2020). Consequently, there is a need to wider the range of potential colonizers to assess the full potential of bacterial migration through bioplastics and special bacteria accumulation. Thus, in this research, the main 2 experiments will be set *in situ* in an estuary.

The Medway Estuary and Marshes

Whilst there are existing studies on MC and BBP or TP in the environments, very few have been set in an estuarine setting. As previously discussed, estuaries are area of intense pollutant accumulation and physicochemical changes. The represent the interface between terrestrial and marine environments and between fresh and saltwater.

This study will focus on the Medway Estuary and Marshes. This low lying, flat area of 4749 Ha is situated in North Kent. It forms a single tidal system and joins the southern part of the Thames Estuary. The complex arrangement of tidal channels form peninsulas of grazing marsh and salt marshes islands. Together with the large area of mudflats, intertidal-flats, and sea walls intersected by dikes and fleets, the Medway estuary is home to complex and diverse mixes of coastal habitats which makes it the perfect site to set the present research (Natural England accessed on 05.12.2020). It has also been designated a biological Site of Special Scientific Interest and a Special Protected Area for its variety of habitats and internationally important wintering, passage and breeding birds ((*SSSI* Natural England - accessed 05.12.2020).

In addition, it will work better in collaboration with the Interreg project.

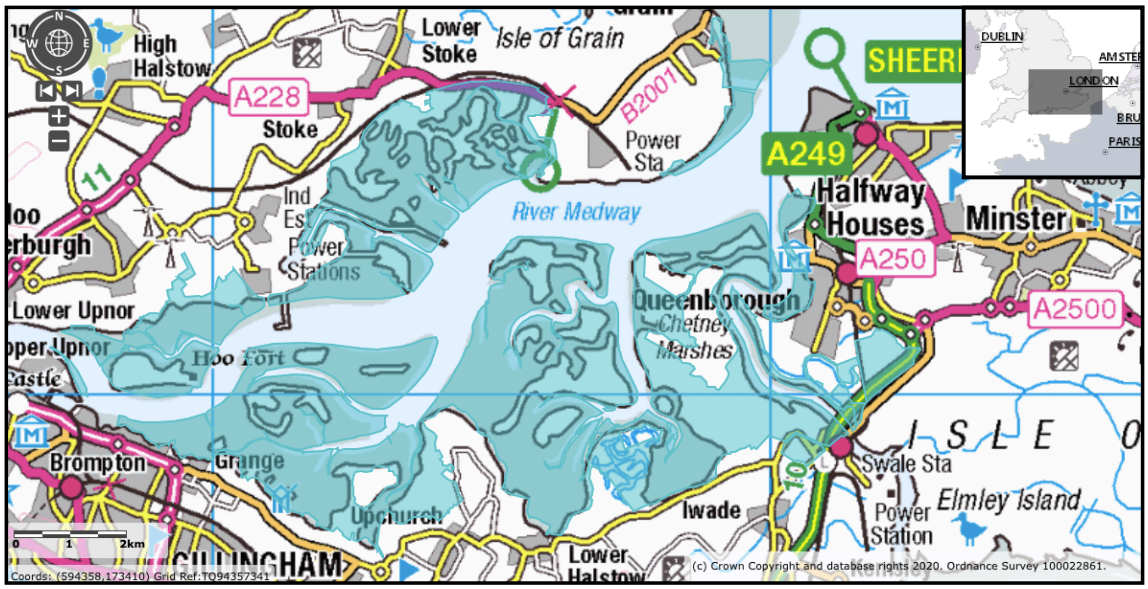

Figure 1 Map of the Medway Estuary retrieved from DEFRA on 06.12.2020.

https://magic.defra.gov.uk/MagicMap.aspx?chosenLayers=vmlIndex,25kIndex,50kIndex,250kIndex,miniscaleIndex,vmlBWIndex,25kBWIndex,50kBWIndex,250kBWIndex,miniscaleBWIndex&box=566851:161502:609080:186419&useDefaultbac

Time

Although the MC dynamics depend on the habitat and matrix, time plays a major role in structuring MCs. MC temporal dynamic and degradation can go through stages. Some bacteria might start degrading the plastic and ten completely disappear from the plastic and another genra will form a biofilm (De Tender *et al*., 2017). Long term monitored studies are important to understand the complete picture of the colonisation succession, dynamic, their changing function and biodegradation. In addition, extensive and deep characterizations of communities associated to BBP using modern techniques such as high-throughput sequencing methods have only been done on a short timescale and one type of BBP at a time (Dussud *et al*., 2018b; Pinnell and Turner, 2019). Moreover, given the highly dynamic nature of estuaries and microbiota seasonal and weather response, it is very likely that samples will vary considerably between sampling times (Vermeiren, C. Munoz and Ikejima, 2016). It is therefore important to sample regularly, at least at each seasonal change and over a year.

Types of plastics

To date, most studies have focussed on 1 or 2 types of bioplastic at a time. It would be wise to compare the most produced/used ones.

Based on their production and uses, Starch blends, Polylactic acid (PLA), polybutylene adipate terephthalate (PBAT) and the polyhydroxyalkanoates (PHAs) will be considered for the experiments. Their production is estimated to intensify substantially and as a result are at increasing risk of being inappropriately managed and released into aquatic environments. In addition, BBPs will be chosen in function of the types used by other BIOPeP PhD students for comparative purposes and subject to availability with BIOPeP project partners.

4.2. Methods

Fieldwork 1 (F1): O 1, 3 & 4: Increase a baseline of the microbial communities' diversity and distribution associated with TP considering environmental physicochemical factors within estuaries. Determine biological processes involved in their transport and fate.

The first objective will be the baseline knowledge on TP for later comparison with BBP results. The idea is to sample plastics and surrounding water and sediment found in the Medway Estuary. Various sampling location will be determined along the salinity gradient, the oxygen level and different habitats (e.g.: salt or grazing marsh, mudflat, sea walls).

Then, in function of the different physicochemical factors:

1. Look at the occurrence and distribution of plastics across the Medway Estuary. (**O1**)
2. Determine biofouling and biodegradation involvement in transport/fate. (**O3**)
3. Characterise TPs, sediment and water microbial communities. (**O1**)
4. Characterise enzymatic activity (**O4**)

Fieldwork 2 (F2): Objectives 2, 3, 4, 5 & 6: Understand MC and BBP interaction to determine the processes involves in their transport and fate, whether and how they biodegrade, carry special bacteria and alter biogeochemical functions of different ecosystems along the estuary. For these objectives, different types of BBP, TP and ceramic (for control (Pinnell and Turner, 2019)) will be put in glass fibre bags and exposed along the estuary. The sampling locations will be the same than for the first experiment. BBP and TP biodegradation will be monitored. Attached BBP, TP and surrounding water and sediment microbial communities will also be monitored. Then:

1. Evaluate biodegradation and biofouling (**O3, O5**)
2. Characterise BBP, sediment and water microbial communities. (**O2, O4, O5, O6**)
3. Characterise enzymatic activity (**O4, O6**)

Side project

Danielle Marchant (Interreg), Gabriela Zemelka (Interreg) and Ana Martinez Rodriguez (BIOPeP) have set up 30 mesocosms at the River Laboratory in Dorset (Figure 2). The ponds are inoculated until spring with freshwater from a nearby river (pumped and siphoned), sediments (95% building sand + 5% mushroom compost, 3.5cm deep), phyto-zooplanktons and macroinvertebrates (scrapers/ grazers, shredders, collector/ gatherers, collector – filterers, predators, other – omnivores, from a riverine source).

In spring, three plastic treatments will be added: BBP, TP, combination of both. Each plastic treatment will receive two levels of nutrient load. For each treatment and nutrient level, there will be 5 replicates. Temperature, pH and conductivity will be monitored weekly. Types of BBP and TP are yet to be determined.

As a side project, samples of the microbiota from the plastic, the sediments and the water column will be taken regularly (once a month?), as stipulated in the sampling protocol section.

This will help understand the microbiota biogeochemical function considering nutrient load and plastic type along with microbial community temporal dynamic and structure in comparison with traditional plastic. It will also look at the influence of presence of traditional plastic on BBP MC.

Figure 2 Mesocosms set up at the River Laboratory in Dorset

4.3. Research Techniques

F1 & 2: The physicochemical characteristics at each sampling locations will be measured. Underwater logger oxygen sensor (Unisens type) will quantify oxygen levels. Portable multi-meter (Hach HQ40D type) will assess pH, conductivity and temperature. Substrate and vegetation cover will be recorded with the 0.5x0.5 metal quadrat. For nutrient levels, water will be filtered through a 0.2 µm filter membrane into a gas type vile with Zinc Chloride, and processed through an automated Skalar in laboratory. Depth and salinity will also be measured.

F1: Plastic macroparticles and microparticles can be retrieved at each location using a variety of techniques: using forceps, 37.5- 48 µm sieves or filter membranes, density flotation (to extract microplastics from sediment samples), mesh nets.

The plastic sizes, colours and shapes can be determined by the naked eye or using a microscope and their type using Fourier-Transformed Infrared Spectroscopy (FTIR) or a set of density analyses (Barnett *et al*., 2020).

F1&2: Microbial communities can be characterised by first extracting the DNA from sediments, water 0.2μm membrane filters or plastic samples (using soil and water DNA kits). Then, Polymerase Chain Reactions (PCR) can be applied to amplify hypervariable regions of bacterial and archaeal 16SrRNA genes. The product can then be sequenced by Next Generation Sequencing/High throughput sequencing (e.g.: Illumina HiSeq).

BBP and TP biodegradation and biofilm content can be determined using a set of Buoyancy experiments (Wu, Zhang, Li, *et al*., 2020), optical density techniques (Li *et al*., 2019) and Scanning Electron Microscopy (SEM).

Enzymatic activity can be assessed by extracting RNA and transcriptomic/gene pool analyses; however, more techniques need to be found.

Biogeochemical functions can also be assessed through further laboratory experiments.

5. Data analysis indications

Using bioinformatics, the obtained sequences from the hypervariable region of the 16S rRNA gene will be filtered for quality, aligned and their phylogenetic trees produced. Useful genetic analysis software are MEGA, Geneious, MAAFT, Bioedit.

Specie taxonomic annotation of the sequences can then be done using the Silva Database.

In all studies, further analyses such as calculation of core communities between samples, heatmaps and bar plots are conducted using the Quantitative Insights Into Microbial Ecology (QIIME) and R software packages.

The microbial communities obtained will also be compared with libraries of known bioplastic degrading bacteria such as the Comprehensive Plastic Microbial Biodegradation Database or the PHA DE Database. However, it is very likely that new BBP-degrading bacteria will be discovered with the help of transcriptomic analyses.

There will be opportunities to compare findings with other BIOPeP PhD students.

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Risks

Risk	Impact	Likelihood	Mitigation
Unable to do fieldwork in the Medway estuary	Delay on all experiments	Moderate (COVID, special permits delivery delays)	Obtain permits as soon as possible. Find alternative study site, closer to London for easy travel.
Unable to finish my experiments as planned	Reduces the data analysis period. Unable to troubleshoot in case something is not right. Incomplete dataset. Unable to reach all research objectives.	Moderate (COVID, Lockdown)	Start fieldwork ASAP. Will likely have enough to analyse regardless.
Being overwhelmed by the amount of data to analyse.	Data badly analysed. Data analysis taking too long. Delays in the thesis write up.	High (many factors tested, many objectives, many samples and DNA sequences)	Have a solid data analysis and data sorting plan. Ask for advice. Read.
Personal computer power limitation for data analysis	Data analysis taking too long	High (Old computer, data analysis requiring high computer power)	Look for QMUL computational capacity and plan when to use in advance.

Record of postgraduate training and personal development plan

Training needs:

- FTIR: TBD
- Scanning Electron Microscopy: TBD
- DNA and RNA extraction refresh: TBD, conducted by Dr Chloe Economou
- JP Building induction: early January 2021

The Continuing Personal Development, Skill Point System record and Personal Development Plan can be found below.